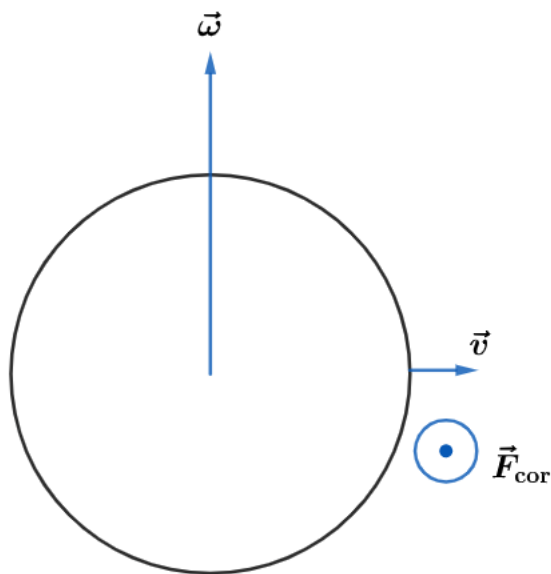


2021 F=ma Exam: Problem 4

Kevin S. Huang



In Bethany's frame (rotating frame), Alice gets deflected to the west by the Coriolis force $\vec{F}_{\text{Cor}} = 2m\vec{v} \times \vec{\omega}$. From kinematics, the amount of deflection d scales as

$$d \propto at^2$$

We have

$$a \propto F_{\text{Cor}} \propto v \propto \sqrt{h}$$

using conservation of energy ($\frac{1}{2}mv^2 = mgh$). For free-fall motion, we know $h \propto t^2$:

$$d \propto \sqrt{h} \cdot h = h^{3/2}$$

Thus,

$$\begin{aligned} \frac{D'}{D} &= \frac{(2h)^{3/2}}{h^{3/2}} \\ D' &= 2^{3/2}D \end{aligned}$$

so the answer is E.